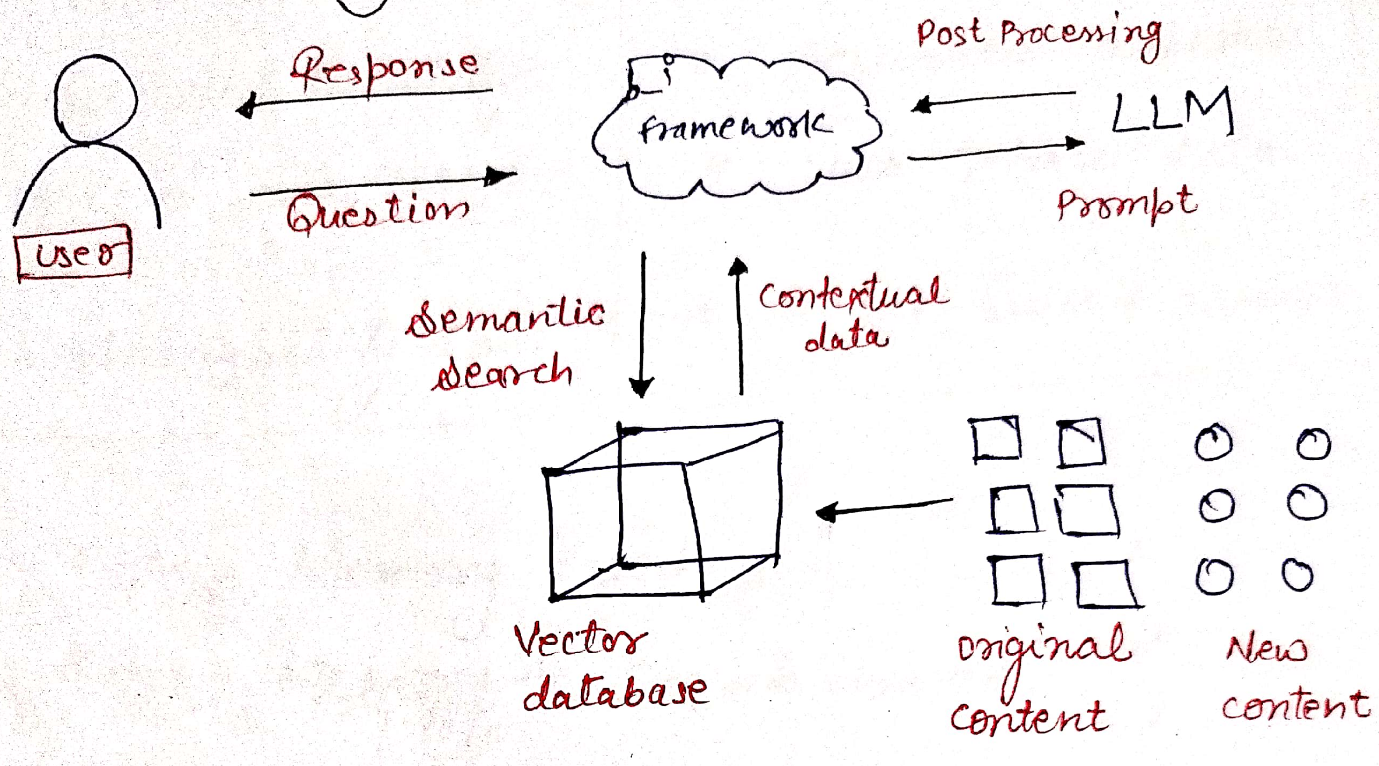
Rag architecture.



What is RAG?

- Retrieval → Finding information.
- Augmented → Adding info with context
- Generation → Generating final answer using LLM.

* It works with vector db & LLM
* It provides the private data access to LLM

Traditional LLM	RAG
* Hallucinate	* Real time data use
* Outdated knowledge	* Accurate answers.
* Private data access - No.	* Uses private data
* Explainability Low.	* Source based answer.

LLM = Memory based.

RAG = Memory + Search based.

Complete RAG pipeline.

1. Data collection → pdfs, Documents, websites, databases.
2. chunking. → Breaking large document into small chunks.
3. Embedding → converting text into numbers (vectors)
4. Vector Db → stores embedding (FAISS, Weaviate, Pinecone)
5. Query-input → "What is Company leave policy?"
6. Retrieval. → Finding relevant chunks (Similarity search)
7. Augmentation → Retrieved data + Query query.
8. Generation → LLM Generates final answer.

If you want to create Rag pipeline.

Pillar 1 :- Embedding Model →
- OpenAI ~~tm~~ text-embedding-ada-002.
- open source models of Hugging face.

Pillar 2 :- Vector database →
- pinecone
- chromaDB
- Weaviate.

Pillar 3 :- LLM (Language Model) →
- GPT-4
- Claude
- Gemini

Pillar 4 :- Orchestration Framework →
- Langchain
- LlamaIndex.

Pillar 5 :- Chunking strategy →
- Fixed Sum
- semantic chunking.
- sliding window.

Real Cases

* Customer Support chatbot
* Legal Document search.
* Medical Assistant
* Code Assistant
* HR & Internal Tools.

Traditional LLM →	~~pr~~ closed book, outdated, hallucinate.
RAG →	open book, real-time, Grounded answer.